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HighFidelityEphemerisModel.jl: A Julia package for high-fidelity astrodynamics

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Summary

HighFidelityEphemerisModel.jl is an open-source Julia (Bezanson et al., 2017) package for modeling spacecraft motion in high-fidelity, time-dependent dynamical environments. The package provides equations of motion that account for the gravitational influence of multiple celestial bodies using ephemeris data, together with configurable perturbations from spherical harmonics, solar radiation pressure, and atmospheric drag. Rather than implementing numerical integration algorithms, the package provides dynamical models compatible with Julia's OrdinaryDiffEq.jl ecosystem (Rackauckas & Nie, 2017).

The package supports trajectory propagation in Cartesian coordinates and Gauss variational equations formulated in modified equinoctial, classical Keplerian, and ordinary equinoctial elements. Additional capabilities include state-transition matrix propagation, Jacobian and Hessian evaluation, event-condition utilities, and export of propagated trajectories to SPICE kernels. Ephemeris and reference-frame evaluations are supported through NASA's SPICE toolkit (Acton, 1996) and Julia-native alternatives. These capabilities enable researchers to incorporate high-fidelity spacecraft dynamics into trajectory analysis, sensitivity analysis, and optimization workflows, including applications in cislunar space.

Statement of need

Accurate spacecraft trajectory propagation is essential for mission design, trajectory optimization, and dynamical analysis. Simplified models, such as the two-body problem and circular restricted three-body problem, provide valuable insight into spacecraft motion but may be insufficient for applications requiring higher physical fidelity. In multi-body environments, particularly cislunar space, trajectory propagation may require time-dependent planetary and lunar ephemerides, nonspherical gravitational fields, and non-gravitational perturbations such as solar radiation pressure. Incorporating these effects requires consistent treatment of celestial-body positions, reference-frame transformations, and force models.

Developing and validating these dynamical models independently can require substantial effort, especially when they must be repeatedly evaluated within numerical optimization and sensitivity-analysis workflows. While numerical integration libraries provide algorithms for solving differential equations, astrodynamics applications additionally require specialized equations of motion, perturbation models, and derivative information. Providing these components through reusable software can reduce duplicated implementation effort and facilitate reproducible numerical experiments.

HighFidelityEphemerisModel.jl addresses this need by providing configurable, ephemeris-based spacecraft dynamics within Julia's scientific computing ecosystem. By separating the equations of motion from the numerical integration algorithms, the package enables researchers to combine

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high-fidelity dynamical models with existing differential-equation solvers and optimization tools. It is intended for researchers and engineers developing spacecraft trajectory analysis, design, and optimization methods, particularly those requiring direct programmatic access to the underlying dynamics.

State of the field

Several open-source software packages provide high-fidelity spacecraft dynamics and trajectory propagation capabilities. Orekit (Maisonobe et al., 2010) is a Java-based flight dynamics library offering comprehensive orbital propagation, reference-frame transformations, and mission analysis functionality. NASA's General Mission Analysis Tool (GMAT) (Hughes et al., 2014) provides an integrated environment for trajectory design, optimization, and navigation through graphical, scripting, and programmatic interfaces. Basilisk (Kenneally et al., 2020) is a modular simulation framework that combines spacecraft dynamics with guidance, navigation, and control models. Nyx (Nyx Space, 2026) provides a Rust-based astrodynamics toolkit for trajectory propagation and orbit determination, while Brahe (Eddy & Kochenderfer, 2026) offers a composable astrodynamics library with Rust and Python interfaces, particularly for satellite dynamics and mission operations.

Although these tools offer extensive capabilities, HighFidelityEphemerisModel.jl addresses a more focused requirement: providing configurable, ephemeris-based equations of motion that can be directly incorporated into Julia's scientific computing workflows. Rather than providing a comprehensive mission analysis environment or implementing separate numerical integration algorithms, HFEM exposes reusable dynamical models compatible with OrdinaryDiffEq.jl. This enables researchers to combine high-fidelity force models with numerical integration, sensitivity analysis, and trajectory optimization methods within the same programming environment.

Within the Julia ecosystem, `SatelliteToolbox.jl` offers satellite-analysis tools, including analytical orbit propagators and terrestrial reference-frame utilities (JuliaSpace contributors, 2026). `GeneralAstrodynamics.jl` provides common astrodynamics calculations and solvers for Keplerian, Lambert, and halo-orbit problems (JuliaAstro contributors, 2026). `AstroModels.jl` develops reusable gravitational and other astrodynamical force models (Julia Space Mission Design contributors, 2026a). HFEM complements these projects by packaging ephemeris-based spacecraft equations of motion, multiple ephemeris backends, and Cartesian and Gauss variational formulations for direct use with Julia differential-equation solvers.

Software design

Dynamical model

`HighFidelityEphemerisModel.jl` formulates spacecraft motion relative to a central body using ephemeris-based gravitational dynamics and configurable perturbation models. The Cartesian equations of motion are expressed as

$$\dot{\boldsymbol{x}} = \begin{bmatrix} \boldsymbol{v} \\ -\dfrac{\mu}{\|\boldsymbol{r}\|^3}\boldsymbol{r} + \displaystyle\sum_i \boldsymbol{a}_{\mathrm{3bd},i} + \boldsymbol{a}_{\mathrm{SH}} + \boldsymbol{a}_{\mathrm{SRP}} + \boldsymbol{a}_{\mathrm{drag}} \end{bmatrix},$$

where $\boldsymbol{x} = [\boldsymbol{r}^T, \boldsymbol{v}^T]^T$ is the spacecraft state, μ is the central-body gravitational parameter, and the remaining terms represent third-body gravity, spherical harmonics, solar radiation pressure, and atmospheric drag, respectively. The perturbation terms are configurable, allowing the same framework to represent point-mass N-body dynamics or more detailed gravitational and non-gravitational environments.

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Third-body accelerations are evaluated using ephemeris-derived body positions and Battin's numerically stable formulation (Battin, 1987). Spherical-harmonic accelerations are evaluated in the corresponding body-fixed frame and transformed into the propagation frame. The implementation also supports nondimensional state and time scaling through configurable characteristic distance and time units.

For the prograde modified equinoctial elements (MEE) $\boldsymbol{y} = (p, f, g, h, k, L)$ (Walker et al., 1985), p is the semi-latus rectum and

$$\begin{aligned} f &= e\cos\varpi, & g &= e\sin\varpi, & \varpi &= \Omega + \omega, \\ h &= \tan(i/2)\cos\Omega, & k &= \tan(i/2)\sin\Omega, & L &= \varpi + \nu. \end{aligned}$$

Here $e, i, \Omega, \omega, \nu$ denote eccentricity, inclination, ascending-node longitude, argument of periapsis, and true anomaly. Define $w = 1 + f\cos L + g\sin L$, $s^2 = 1 + h^2 + k^2$, $q = \sqrt{p/\mu}$, and $\chi = h\sin L - k\cos L$. The implemented Gauss variational equations are

$$\begin{aligned} \dot{p} &= \frac{2pq}{w} a_T, \\ \dot{f} &= q\left[\sin L\, a_R + \frac{(w+1)\cos L + f}{w} a_T - \frac{g\chi}{w} a_N\right], \\ \dot{g} &= q\left[-\cos L\, a_R + \frac{(w+1)\sin L + g}{w} a_T + \frac{f\chi}{w} a_N\right], \\ \dot{h} &= \frac{qs^2\cos L}{2w} a_N, \qquad \dot{k} = \frac{qs^2\sin L}{2w} a_N, \\ \dot{L} &= \sqrt{\mu p}\left(\frac{w}{p}\right)^2 + \frac{q\chi}{w} a_N. \end{aligned}$$

The components (a_R, a_T, a_N) contain only perturbing acceleration: radial outward, transverse along $\hat{\boldsymbol{N}} \times \hat{\boldsymbol{R}}$, and normal along $\boldsymbol{r} \times \boldsymbol{v}$. HFEM subtracts central two-body acceleration before this projection; the first term in $\dot{L}$ supplies Keplerian drift. Consistent physical or canonical units are required, with μ the corresponding central-body parameter and angles in radians.

Modular implementation

A central architectural decision is the separation of dynamical models from ephemeris evaluation and numerical integration. HFEM provides three parameter backends: `SpiceParameters`, which queries NAIF SPICE directly (Acton, 1996); `EphemeridesParameters`, which uses the Julia-native `Ephemerides.jl` and `FrameTransformations.jl` libraries (Julia Space Mission Design contributors, 2026b, 2026c); and `InterpParameters`, which evaluates pre-interpolated ephemeris and transformation data.

Julia's multiple dispatch allows the same high-level equations-of-motion functions, `eom_Nbody!` and `eom_NbodySH!`, to operate with these different parameter types. Direct SPICE evaluation provides an established reference implementation, while the Julia-native backend supports automatic differentiation and tested multithreaded propagation workflows without relying on direct CSPICE calls during each force evaluation. The interpolated backend is retained for compatibility, although its accuracy depends on the interpolation resolution.

Numerical integration is delegated to `OrdinaryDiffEq.jl` (Rackauckas & Nie, 2017). This self-contained example propagates a Moon-centered state with Earth and Sun point-mass perturbations. Run it from the HFEM repository using `julia --project=.` after installing its declared dependencies. Download the required NAIF kernels beforehand and set the `SPICE` environment variable to their root directory, with `lsk/`, `spk/`, and `pck/` subdirectories.

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```julia
using HighFidelityEphemerisModel
using OrdinaryDiffEq
using SPICE

spice_dir = ENV["SPICE"]
furnsh(joinpath(spice_dir, "lsk", "naif0012.tls"))
furnsh(joinpath(spice_dir, "spk", "de440.bsp"))
furnsh(joinpath(spice_dir, "pck", "gm_de440.tpc"))

ids = ["301", "399", "10"]  # Moon first, then Earth and Sun
GMs = [bodvrd(id, "GM", 1)[1] for id in ids]  # km^3/s^2
et0 = str2et("2026-01-05T00:00:00")  # UTC input
DU = 1e5  # km
params = SpiceParameters(et0, DU, GMs, ids, "J2000", "NONE")

x0 = [1.05, 0.0, 0.3, 0.5, 1.0, 0.0]
tspan = (0.0, 6 * 3600 / params.TU)
prob = ODEProblem(eom_Nbody!, x0, tspan, params)
sol = solve(prob, Vern8(); reltol=1e-12, abstol=1e-12)
@assert sol.t[end] == tspan[2] && all(u -> all(isfinite, u), sol.u)
kclear()
```

The state and time use canonical units: $\mathrm{VU} = \sqrt{\mathrm{GM}_{\mathrm{Moon}}/\mathrm{DU}}$ and $\mathrm{TU} = \mathrm{DU}/\mathrm{VU}$. The six-hour example omits harmonics, SRP, and drag; solver choice remains independent of the force model.

Figure 1 summarizes these interfaces.

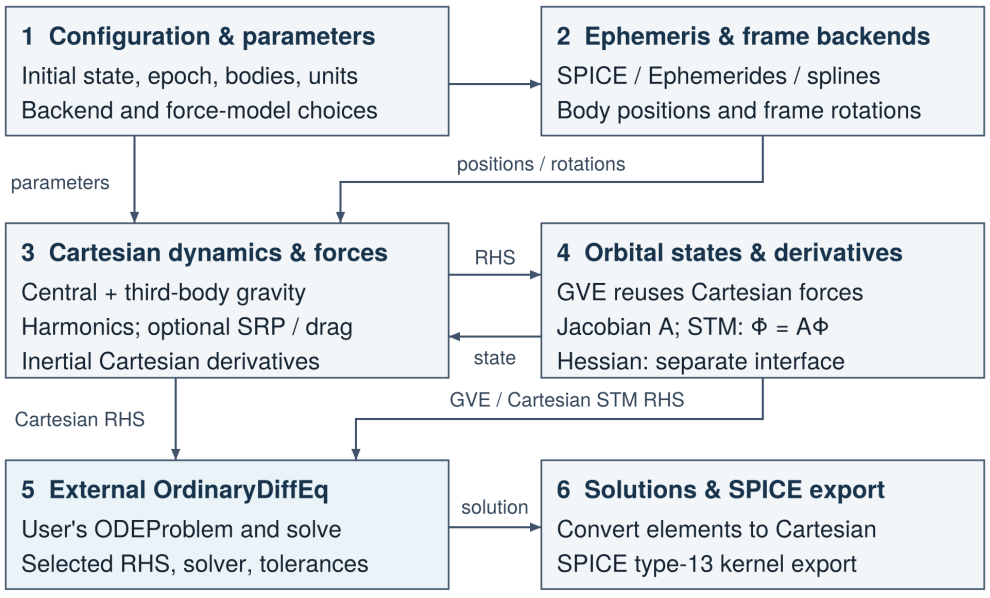


Figure 1: HFEM architecture. Cartesian forces support direct propagation and Gauss variational equations (GVE). Jacobians drive Cartesian state-transition matrix (STM) evolution; Hessians are separate. Element solutions require Cartesian conversion before SPICE export.

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Orbital representations and research interfaces

In addition to the MEE formulation described above, HFEM supports Gauss variational equations in classical Keplerian and ordinary equinoctial elements. Orbital-element conversions are provided through `AstrodynamicsCore.jl` (eXplorationLogisticsControl, 2026).

Prograde MEE support elliptic, parabolic, and hyperbolic trajectories for $p > 0$ and $w \neq 0$, but are singular at $i = \pi$. Classical Keplerian elements are singular for circular and equatorial orbits and ill-conditioned near the parabolic boundary. The implemented ordinary equinoctial formulation is restricted to elliptic trajectories.

The package also provides automatic-differentiation-based Jacobian and Hessian calculations through `ForwardDiff.jl` (Revels et al., 2016), together with state-transition matrix propagation for sensitivity analysis. Propagated trajectories can be exported to SPICE type-13 ephemeris kernels, with utilities supporting multiple trajectory segments and maneuver metadata. Together, these capabilities facilitate the integration of high-fidelity spacecraft dynamics into trajectory optimization, guidance, and reproducible mission-analysis workflows.

Research impact statement

`HighFidelityEphemerisModel.jl` has supported research on high-fidelity trajectory design and station keeping in the Earth–Moon system (Phung & Shimane, 2026). This application demonstrates the integration of ephemeris-based spacecraft dynamics, sensitivity analysis, and trajectory propagation into larger numerical optimization and mission-analysis workflows.

The package has also been evaluated through numerical comparisons with established astrodynamics software, including GMAT, Orekit, Nyx/ANISE, and Basilisk. Three representative cases encompass a seven-day near-rectilinear halo orbit (H7), a low lunar orbit (LLO-MESH4), and a low Earth orbit (LEO-EMSH4). Each case includes third-body gravitational perturbations and degree/order-four spherical harmonics. Residuals use documented HFEM SPICE trajectories, with tool-specific numerical mirrors for Nyx/ANISE and LLO Orekit; they measure agreement, not absolute accuracy. These comparisons provide quantitative evidence of numerical consistency across different propagation frameworks while accounting for differences in integrators, ephemeris evaluation, and model implementation.

Table 1, Figure 2, and Table 2 summarize these comparisons. GMAT H7 uses shared MOON_PA orientation with an inertial-frame correction. GMAT LLO/LEO endpoint offsets from nominal grids reach 2.70 microseconds. Basilisk uses 1 s communication/task updates with adaptive integration; Orekit uses shared frame orientations.

Table 1: Validation cases; frames specify harmonic evaluation axes. LLO/LEO each span one period.

Case	Central body	Third bodies	4x4 frame	Span
H7	Moon	Earth + Sun	MOON_PA	7 d
LLO-MESH4	Moon	Earth + Sun	MOON_PA	2.064 h
LEO-EMSH4	Earth	Moon + Sun	IAU_EARTH	1.978 h

Table 2: Maximum sampled position residuals against documented HFEM SPICE references or numerical mirrors, not absolute trajectory errors.

Implementation	H7 [mm]	LLO-MESH4 [mm]	LEO-EMSH4 [mm]
HFEM Ephemerides	0.337568	0.0349237	0.148159
GMAT	136.282	3.07436	32.1019
Orekit	5.31215	1.51733e-05	0.000104418

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Implementation	H7 [mm]	LLO-MESH4 [mm]	LEO-EMSH4 [mm]
Nyx/ANISE	0.00121614	0.221886	1.02998
Basilisk	0.347638	0.0262097	101.602

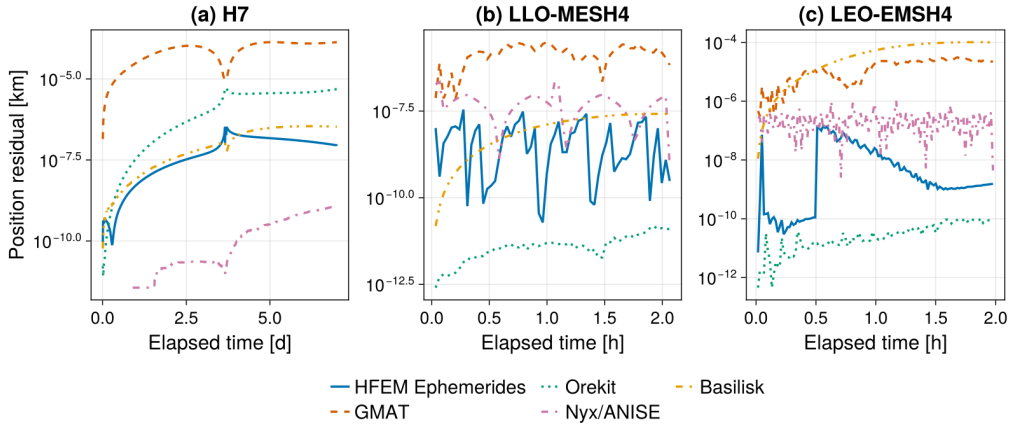


Figure 2: Position residuals against HFEM SPICE trajectories or tool-specific numerical mirrors, not absolute trajectory accuracy. Initialization ($t = 0$) is omitted; later stored zeros appear as logarithmic gaps without flooring. Output sampling differs from integration steps.

Additional benchmarks evaluate HFEM's alternative ephemeris backends. The Julia-native `Ephemerides.jl` backend reduced propagation time relative to direct SPICE evaluation from 0.2473 to 0.1236 s for a point-mass near-rectilinear halo orbit case, and from 0.8677 to 0.7031 s with lunar spherical harmonics. These timings are configuration-specific, not universal speedups.

Table 3 reports end-to-end propagation at HFEM source commit `2aca3a70ffffad4e61db28f68c8c9d6d763` (Julia 1.10.11). For each case/backend, an initial untimed solve excludes the initial Julia compilation run; a separate allocation-measurement solve also precedes the ten timed repetitions. No compilation time is subtracted from measured runs. Timings include ODE construction, Vern8 propagation and in-memory output with initialized backends; backend setup and serialization are excluded.

Table 3: Median propagation (ten repetitions, one Julia thread). Ratio: Ephemerides/SPICE. P3 uses point masses; H7 adds lunar 4x4.

Case	SPICE [s]	Ephemerides [s]	Time ratio
P3	0.2472955	0.1236207	0.4999
H7	0.8677467	0.7031303	0.8103

The package is distributed through Julia's package registry and provides documented examples, automated tests, and continuous integration to support reproducible research and further software development.

AI usage disclosure

AI was used to help develop the code and to draft and edit parts of this manuscript. All text produced with AI assistance was reviewed, revised, and verified by the authors before inclusion in the code or paper.

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